EVOQUA WATER SYSTEM
80 GPM 2-PASS RO CONTAINER

ELECTRIC HYDROGEN – V1 HYDROGEN PLANT, NATICK, MA C/O M&H CONSULTING

Budgetary Proposal # 587118 Rev A

June 23, 2023

210 Sixth Avenue, Suite 3300, Pittsburgh, PA 15222
(866) 926-8420 (toll-free)   (978) 614-7111 (toll)   www.evoqua.com

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1.0 Equipment Scope

1.1 Equipment

The list below details the unit operations supplied as part of the Evoqua Scope of Supply

Item #	Qty	Description
1.	1	**40 Foot standard shipping Container includes:** • Exterior painted and interior Insulated • Outfitted with Lights • Steel diamond floor plate • Heating/AC unit installed • Power Distribution Panel • PLC with 9" HMI for comfort control • Ethernet Communication and remote Monitoring
Items below mounted in Container from item 1. Interconnecting Piping and Valves in PVC (see P&ID for Details)		
2.	1	**Pre-treatment Cartridge Filter Housing** • Inlet/Outlet/Bypass/ and Vent Valves • Inlet/Outlet Pressure Gauges • 304SS construction • Contains 21 – 30" long elements
3.	2	FRP 42 cubic foot - Activated Carbon Service exchange Vessels (to be supplied under a separate service agreement)
4.	1	**Antiscalant Addition** • Fed from chemical tote • One (1) Injection pump • One (1) low level switch
5.	1	**pH Adjustment (acid or caustic** • Fed from chemical tote • One (1) Injection pump • One (1) Low level switch • pH cell and Signet 8900 series monitor located on RO skid. • Chemicals to be injected will be based on feedwater chemistry

Item #	Qty	Description
6.	1	**Vantage Two Pass RO with Caustic interstage Injection- M284R044DMAD** Includes: • 80 GPM product • Includes: • VFD controlled booster pump • 5-micron pre-filter –Protects the RO membranes from particulate contamination • Two Pass RO – 4:2:1 // 2:1:1 array with 8" diameter 4M RO vessels. • CPA5-34 in the first pass, ESPA2-MAX in the second pass • 8" diameter FRP RO pressure vessels. • Vessels are ASME code rated to 450 psi with the RP-1 stamp. • Caustic Injection – to reduce feed water CO2 to downstream mixed bed to reduce operating cost • Painted carbon steel Frame – Glass bead Blast • NEMA 4 Power/Control Enclosure • 316SS High Pressure Piping • PVC Low Pressure Feed and Product piping • See the attached specification for more details.
7.	1	**TOC reduction Ultraviolet Sterilizer** • UV Intensity Meter • Recommended to be in recirculation stream from storage tank for continuous TOC reduction
8.	2	FRP 42 cubic feet Service mixed bed ion exchange Vessels (to be supplied under a separate service agreement)

1.2 On Site Services

In addition to the equipment above, these services are offered as an option. Pricing will be based on location and to be service by nearest Evoqua service organization:

Qty.	Description
To Follow	On-site (10 hrs/day; 1 trip) for Installation Support & Inspection
To Follow	On-site (10 hrs/day; 1 trip) for Start-up Support & Assistance
To Follow	On-site (10 hrs/day; 1 trip) for Hands-on Training / Instruction, using Evoqua system O&M manual based on Level One training.
Note	Travel & Living expenses will be invoiced per the enclosed Field Service Policy, and additional days of service can be purchased, according to the enclosed Field Service Policy (see Section 3.3 for further details). “Repair or replacement of parts” under warranty terms, is repair of the part or a new part, not including removal and installation.

Estimates of required time do not include any mandated attendance at site safety training or other site requirements.

Evoqua Services & Products

Evoqua Water Technologies LLC Service & Products organization consists of over 80 branches throughout the United States, Puerto Rico, and Canada. 85% of the US population lives within 2 hours of an Evoqua Branch. Each Evoqua Branch is focused on helping you service your process water systems over time. Proper care and maintenance over time is the very best way to minimize your risk of down time while lowering the life cycle cost of your system.

Evoqua is located locally so we can offer you the very best service response time in the industry. Response time can literally mean the difference between having water and not having water within your manufacturing and lab environments. Evoqua's local branches and highly trained service managers and technicians can address and fix problems with your systems within hours. When service is not local, this can sometimes turn into days. Additionally, each Evoqua Branch routinely stocks many of the items used in your system, providing yet another advantage when speed of repair is needed.

Evoqua offers a broad line of products and services that address your water system needs. Examples of these are as follows

Products	Services
Filters & Housings	Preventative Maintenance Contracts
Gauges, Pumps, Controls, Valves & Gaskets	Membrane Care Program
UV Bulbs	Service Deionization
Sterilants	Mobil Water Treatment
Resins & Media	Design, Build, Own, Operate – Evoqua
Membranes (RO, UF, MF and ceramic)	Water Analysis
Lab Systems and Cartridges	Wastewater Ion Exchange Services

Original Manufacturers Parts	
SOPs – Standard Operating Procedures	
System Spare Parts	

During project execution, Evoqua will work with you to tailor a service program that meets your needs. At the time of the Factory Acceptance Test a detailed Preventative Maintenance Proposal will be presented. With Evoqua knowledge of water and your system we are well positioned to provide system care over time that will minimize risk of down time while reducing life cycle ownership cost. Additional detailed information is always available through your local Evoqua Service & Products Representative.

1.3 Pricing Information

To:	ELECTRIC HYDROGEN	Reference:	Proposal # 587118 Rev. A
	1 STRATHMORE ROAD	Proposal Date:	June 23, 2023
	NATICK, MA		Evoqua Water Technologies LLC
	C/O M&H CONSULTING	PO Remittance	210 Sixth Avenue, Suite 3300
		Address:	Pittsburgh, PA 15222
			719-570-9600

Evoqua proposes to furnish materials, equipment and/or technical service in accordance with this quotation**.** Materials, equipment and/or technical service not specifically listed under the Scope of Supply or on attachments to this proposal are EXCLUDED.

Equipment Base Bid Price (2 Units plus first time Engineering)............................. US$1,240,208

Additional Single Units.. US$598,959

PLEASE BE ADVISED THAT THE ENCLOSED BUDGETARY PROPOSAL IS A NON-BINDING DOCUMENT, IS BEING UTILIZED FOR REVIEW AND INFORMATIONAL PURPOSES, AND DOES NOT CONSTITUTE AN OFFER FOR ACCEPTANCE. RECENT MARKET CONDITIONS HAVE RESULTED IN VOLATILITY IN MATERIALS, COMMODITY AND FABRICATION PRICES. THE COSTS USED FOR DEVELOPING THIS BUDGETARY PROPOSAL MAY BE SUBJECT TO ESCALATION. IT SHOULD BE UNDERSTOOD THAT EVOQUA HAS NOT INCLUDED ESCALATION AS PART OF THIS BUDGETARY OFFER AND MAY NEED TO INCLUDE THE ESCALATIONS WHEN FIRM PRICING IS DEVELOPED FOR THIS PROPOSAL IN THE FUTURE.

Freight: FOB Shipping Point, Freight Collect

Shipments: Estimated 40 weeks from acceptance of purchase order and successful T&C negotiations

Submittal Drawings: One (1) set of electronic submittal drawings will be issued based on enclosed document schedule (after receipt and approval of purchase order). Paper Copies of Submittal drawings may be provided as specified and agreed to.

Manuals: One (1) set of English language final prints and O&M manuals on paper, and one (1) electronic copy (bookmarked and linked PDF formatted; for all documents, with the exception of engineering drawings which will be in ACAD).

Terms & Conditions and Payment Terms:
35% due upon acceptance of Purchase Order (PO)
35% due upon submittal of engineering drawings (P&ID)
15% due upon release of fabrication
15% due upon notification of readiness to ship
Net 30

1.4 Optional Supplied Items

In addition to the equipment supplied per Section 1.1 and the quoted options below, Evoqua can supply other optional equipment and services upon request. Pricing for additional items is available upon request and may impact the delivery time. These optional items are detailed in the table below.

Item	Qty	Description	Price (US$)
A			

1.5 Technical Qualifications

The following is a list of Technical specifications that Evoqua has received and reviewed for this proposal. Any requirements not listed here (including "nested" specifications within the attached, but not specifically listed here) have not been considered. Evoqua would be pleased to review any additional requirements and provide a revised proposal considering any resulting impact to price and/or schedule. Clarifications or items to be supplied by others are detailed in the following tables.

Documents

Document Number	Document Name	Rev.	Date	Pages
22MH194-PR8027	ELECTRIC HYDROGEN PRODUCT V1 HYDROGEN PLANT REQUEST FOR QUOTATION WATER PURIFICATION SYSTEM	0	MAY 25, 2023	45

Technical Qualifications

Document Number	Section	Technical Qualification
All	All	Evoqua’s budgetary offer is based on our standard products which provide the quality and quantity of water set forth in Section 2 of this proposal.

2.0 Basis of Design

2.1 Design Requirements

This section details the basis of design for the equipment (the “System”) offered by Evoqua to Electric Hydrogen (the “Purchaser” and “End User/Owner” for the V1 Hydrogen Plant Project in Natick, MA]. Evoqua has used the design information provided in the RFQ 22MH194 dated May 16, 2023, and as listed below to develop our treatment approach, specify and size equipment. Any deviations to the design information contained therein is set forth in Section 1.5, Technical Qualification herein.

The System as fully described in this proposal is based upon the design criteria and design assumptions stated in Table 2.1. Please note that any assumed values are indicated with an asterisk (*). In the event any of the assumptions are incorrect, the proposed System could be affected technically and commercially, therefore we request your confirmation concerning the accuracy of our assumptions.

Table 2.1 Design Criteria and Assumptions	
Influent Water Source	Potable Municipal Surface
Influent Water Pressure: Minimum Maximum	 70 psig 100 psig
Influent Water Temperature: Minimum Maximum	 45 degrees F 90 degrees F
Influent Water Flow Rate: Minimum Maximum	 107 gpm 110 gpm
Instrument Air: Pressure Temperature Dew Point	 80-100psig 100 degrees F -40 degrees F
Chemical Concentrations: Sodium Hypochlorite Caustic Antiscalant Reverse Osmosis Cleaning Solutions	 10 to 12.5% 50% Neat Various
Incoming Power: Controls Pumps	 120-Volt, 1-Phase, 60-Hertz 480-Volt, 3-Phase, 60-Hertz
Seismic Zone	Not applicable
Location	Indoors
Ambient Air Temperature: Indoors - Minimum / Maximum	 40 degrees F / 90 degrees F
Relative Humidity: Indoors - Minimum / Maximum	 TBD
Plant Elevation (Above Mean Sea Level)	Not provided, Various

Table 2.1 Design Criteria and Assumptions	
Wind Load	Not applicable
Snow Load	Not applicable
Area Classification	Unclassified, Non-Hazardous
Notes: * Assumed Value	

2.2 Design Influent Water Characteristics

The Purchaser must ensure that all conditions set forth in this Section, including all the influent water parameters listed in Table 2.2, are met (the "Influent Characteristics").

Table 2.2. Influent Characteristics		
Parameter	**Unit**	**Maximum Design Value**
Total Hardness	mg/l as $CaCO_3$	< 145
Total Suspended Solids (TSS)	mg/l	Not Provided
Turbidity	NTU	< 7
Total Dissolved Solids (TDS)	mg/l	< 500
Total Organic Carbon (TOC)	mg/l	< 8
pH	Standard Units	6.5-9.7
Temperature	°F	40-100
Silt Density Index (SDI), 15 minutes		Not Provided
Conductivity	µS/cm	Not Provided
Colloidal Silica	mg/l	0
Color	Pt/Co	Not measured
Residual Chlorine	mg/l	0
Cations		
Ammonium	mg/l as $CaCO_3$	0
Barium	mg/l as $CaCO_3$	< 2
Calcium	mg/l as $CaCO_3$	< 100
Magnesium	mg/l as $CaCO_3$	< 45
Sodium	mg/l as $CaCO_3$	< 175
Potassium	mg/l as $CaCO_3$	< 8.5
Strontium	mg/l as $CaCO_3$	< 0.225
Anions		
Bicarbonate	mg/l as $CaCO_3$	< 300
Carbonate	mg/l as $CaCO_3$	0
Chloride	mg/l as $CaCO_3$	< 250
Sulfate	mg/l as $CaCO_3$	< 250
Nitrate	mg/l as $CaCO_3$	< 10
Fluoride $^-$	mg/l as $CaCO_3$	< 4
Phosphate	mg/l as $CaCO_3$	o
Other Metals		
Aluminum	mg/l	< 1.6
Barium	mg/l	< 2.0
Boron	mg/l	< 1.0
< 1Iron	mg/l	<1.5
Manganese	mg/l	< 0.3
Strontium	mg/l	< 0.225

Table 2.2. Influent Characteristics

Parameter	Unit	Maximum Design Value
Weak Anions		
Reactive Silica, SiO_2	mg/l	< 20
Carbon Dioxide, CO_2	mg/l	< 7

Notes:

Evoqua will want to confirm Water analyses at each location to confirm whether additional treatment is required.

2.3 Estimated Effluent Water Quality

Table 2.3. Estimated Effluent Water Quality Requirements

Parameter	Unit	Estimated Value
Effluent Flow Rate	gpm	80
Effluent Pressure	psig	30
Total Organic Carbon (TOC)	mg/l	< 5 ppb
Resistivity	MegOhm	18.2

NOTWITHSTANDING ANYTHING IN THIS BUDGETARY PROPOSAL TO THE CONTRARY, EVOQUA IS NOT PROVIDING A PROCESS PERFORMANCE GUARANTEE FOR THE EFFLUENT WATER QUALITY OF ANY TYPE ON THE EVOQUA EQUIPMENT PROPOSED IN THIS BUDGETARY OFFER. THE CURRENT EQUIPMENT DESIGN IS BASED UPON SUPPLYING EVOQUA'S STANDARD EQUIPMENT AS IDENTIFIED IN THIS PROPOSAL. EVOQUA MAY PROVIDE, IN A FIRM PROPOSAL, A PERFORMANCE GUARANTEE CONDITIONED ON COMPLETION OF TREATABILITY TESTING AND BASED UPON MUTUALLY AGREED INFLUENT AND EFFLUENT PARAMETERS & INCLUDE MUTUALLY AGREED UPON TESTING PROTOCOLS AND FURTHER LIMITATIONS AND CONDITIONS.

1. Acceptance of Equipment

Upon completion of start-up, the Purchaser will be asked to accept the equipment and release Evoqua personnel. This written acceptance (see Exhibit 2 Customer Acceptance Letter) serves as an acknowledgment that the equipment was brought into satisfactory operation, the customer's operations personnel were fully instructed in its operation, and suitable arrangement were made for correction of any outstanding problems. Acceptance does not, in any way, relieve Evoqua of its responsibilities under the mechanical and performance guarantees, if any.

EXHIBIT 1

SITE READY CHECKLIST

Evoqua Water Technologies (Evoqua) requires completion of this form to facilitate installation/start-up of your equipment. Please review the notice on page 2, verify the readiness of the following items, and return the completed form to Evoqua. Missing, incomplete and/or incorrect responses may result in additional expenses. Your cooperation is greatly appreciated.

Project Name: ____________________ Date: ____________________

Project No.: ____

Customer Name: ____

Mechanical Contractor: ____

Electrical Contractor: ____

Site Ready Checklist

Item Description	Ready	Not Ready	N/A	Comments (Include Source of Info and Date Where Applicable)
General Site Conditions				
Site(s) Accessible and Directions Available	☐	☐	☐	
Control Building(s) Enclosed	☐	☐	☐	
Power Available at All panels	☐	☐	☐	
“Old” Equipment Removed and Disposed	☐	☐	☐	
Mechanical				
Piping and Vessels				
All equipment installed, leveled, piped anchored, and grounded.	☐	☐	☐	
All piping installation completed; inlet, interconnecting and outlet.	☐	☐	☐	
Inlet water supply available at required flows and pressure.	☐	☐	☐	
All interconnecting piping, air lines, and vessels pressure tested as required, and free of leaks.	☐	☐	☐	
Adequate supports and braces provided for all piping	☐	☐	☐	
Air Supply				
Air supply available at required flows and pressure.	☐	☐	☐	
Air lines, including tubing, complete to all enclosures and valves etc.	☐	☐	☐	
Air is of instrument quality; free of oil, water, etc.	☐	☐	☐	
Drains				
All drain connections meet unit back pressure requirements.	☐	☐	☐	
All drain piping installation complete (at least one pipe size larger than connection at units).	☐	☐	☐	
Drain piping tested for leaks where applicable.	☐	☐	☐	
Pumps and Motors				
All pumps utilized by the system grouted in and aligned by a certified millwright.	☐	☐	☐	

Site Ready Checklist				
Item Description	Ready	Not Ready	N/A	Comments (Include Source of Info and Date Where Applicable)
Motor wiring and MCC complete with proper wire size, fuses heaters, etc.	☐	☐	☐	
Pump motors checked for rotation and vibration prior to coupling.	☐	☐	☐	
All pumps coupled and lubricated as per manufacturer's specifications.	☐	☐	☐	
Miscellaneous				
Media and membrane storage to prevent damage by elements, et., direct sun exposure, rain freezing, etc.	☐	☐	☐	
Pretreatment chemicals available.	☐	☐	☐	
Regeneration chemicals available.	☐	☐	☐	
Safety showers/eyewash must be operational to prevent delays.	☐	☐	☐	
Electrical				
Power				
All electrical tie-ins completed, all circuits "rung out"; including all 120 VAC, 480 VAC, neutrals, and grounds to and from field devices.	☐	☐	☐	
Motor wiring and MCC complete with proper wire size, fuses, heaters, etc.	☐	☐	☐	
Controls				
Equipment On-Site, Visually Inspected, Quantities Verified	☐	☐	☐	
All field panel(s) (off skid) Mounted	☐	☐	☐	
All panel power wiring pulled, labeled, and terminated	☐	☐	☐	
All panel power wiring continuity verified to ensure proper termination.	☐	☐	☐	
All Ethernet cable pulled, labeled, and terminated.	☐	☐	☐	
Ethernet cable tested for proper termination	☐	☐	☐	
All Remote I/O communication cable pulled, labeled, and terminated.	☐	☐	☐	
All Remote I/O communication cable continuity tested to ensure proper termination.	☐	☐	☐	
All panel I/O field wiring (off-skid) pulled, labeled, and terminated.	☐	☐	☐	
All panel I/O field wiring (off-skid) continuity verified to ensure proper termination.	☐	☐	☐	
Computer On-Site, placed in Control Room, and ready for final hook up.	☐	☐	☐	
Computer Furniture Placed and Ready	☐	☐	☐	Provided by Other than Evoqua
Separate Protected Circuit for Computer	☐	☐	☐	Provided by Other than Evoqua
Motor Control Centers Installed and Wired	☐	☐	☐	Provided by Other than Evoqua
Instrumentation				
Equipment On-Site, Visually Inspected, Quantities Verified	☐	☐	☐	
All field instrumentation (off skid) wiring pulled, labeled, and terminated	☐	☐	☐	
All field instrumentation (off skid) wiring continuity verified to ensure proper termination.	☐	☐	☐	

Site Ready Checklist				
Item Description	**Ready**	**Not Ready**	**N/A**	**Comments (Include Source of Info and Date Where Applicable)**
Instrument calibration checked by qualified technician and documented by log sheet or sticker on each device.	☐	☐	☐	

Important Notice

A specific amount of time and number of site visits have been allocated to provide installation/start up service on this project. Please verify the readiness of each of the above items and return a signed copy of this form to Evoqua. Customer must initial those items that are not ready. Failure to complete any of the above items before arrival of Evoqua personnel may result in the following:

1) Reserves the right to leave the site and return when items are completed.
2) If additional time and/or expenses are required, Evoqua reserves the right to assess additional charges.

On Day One of the planned start-up, the assigned Evoqua Field Service Engineer Lead will conduct a walkdown of the installed equipment and system against the P&IDs of record. Following the walk-down a conference call will be arranged by the Evoqua Project Manager with the customer and Evoqua Field Service Engineer Lead in attendance. The results of the site walk-down will be reviewed and a decision will be reached as to whether the system installation is complete and accurate or if further installation or corrections are required to allow for start-up to proceed. If not, then the customer will be given the following options:

1) Have the Evoqua Field Service Team leave site and return when the system installation is complete and/or corrections have been made. The timing of this return visit will be subject to the availability of Filed Service resources at the time.
2) Have the Evoqua Field Service Team remain on-site and provide technical advisory assistance to the installer and customer. In this case the days spent in this capacity will be deducted from the original budgeted on-site days and will have to be replaced by change order to the contract.

Approved by (please print full name)	Title	Signature	Date

Evoqua requires the customer to provide a reasonably safe work environment in accordance with applicable OSHA standards including but not limited to 29CFR1910.146, Confined Space & 29CFR1910.147, Lockout/Tagout.

EXHIBIT 2

ACCEPTANCE OF EQUIPMENT

Customer Name: ______________________ Site Location ______________________

Customer PO # ______________________ ______________________

Sales Order #: ______________________ ______________________

Work Completion Date: ______________________ ______________________

Description of Equipment (or work performed):

____

____

____

The undersigned, an authorized representative of ______________________
(Customer Name/Contractor Name)

acknowledges that (describe work performed, installation, or start-up, or repair)

____

____

____

was performed by: ______________________
(Evoqua Field Service Engineer name or Evoqua Contractor Name)

and completed the work to the satisfaction of the Customer/Customer Contractor's representative. The Customer's operating personnel were fully instructed in its operation and maintenance and suitable arrangements were made for correction of outstanding items (if any) listed below:

____

____

____

Customer acceptance of the equipment in no way relieves Evoqua of its responsibilities under the warranties as set forth in the Proposal or Purchase Order or Signed Contract.

Accepted By: (Customer representative) ______________________

Title: ______________________ **Date Accepted**: ______________________

Evoqua Representative ______________________ **Date** ______________________

Email completed form to: Evoqua Project Manager

3.0 Attachments

The following attachments are appended to this proposal:

3.1 Engineering Deliverables List

3.2 Equipment Specifications

- Vantage M284 RO Specification

3.3 Proposed Drawings

- Process & Instrumentation Diagram